MIL-O-2478H
30 September 1983
SUPERSEDING
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19 March 1971

MILITARY SPECIFICATION

OVEN, BAKING AND ROASTING, DECK, TWO-COMPARTMENT, OIL

This document is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a two-compartment, baking and roasting, oil burning oven equipped with a rotary cup, mechanical atomization-type burner.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-F-800	-	Fuel Oil, Diesel
VV-F-815	-	Fuel Oil, Burner

MILITARY

MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank
MIL-R-11337	-	Range, Packaging and Packing Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7310

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of U.S. Military Property

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

UNDERWRITERS' LABORATORIES, INC. (UL)

Standard No. 296 - Oil Burners
Standard No. 343 - Pumps for Oil-Burning Appliances

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Corrosion-Resisting Chromium-Nickel Steel Plate, Sheets, and Strip

A 176 - Corrosion-Resisting Chromium Steel Plate, Sheet, and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.5).

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specified requirements shall be classified a defect. The inspection level

3.2 Standard product. The oven delivered under this document shall be the manufacturer's current commercial product, except for any changes necessary to comply with the requirements of this document.

* 3.3 UL Standards. The oil burner and oil pump shall conform to the applicable requirements of UL Standard 296 and UL Standard 343, respectively. Electrical and mechanical components shall conform to the applicable UL Standards listed in the appendices A of UL standards 296 AND 343.

3.3.1 Certification. Prior to approval of the first article sample, if one is submitted, or prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the oil-burner and oil-pump he proposes to furnish under this document meet the applicable requirements of UL Standard 296 and UL Standard 343, respectively. Electrical components shall conform to applicable UL Standards. Acceptable evidence of meeting UL requirements shall be the UL listing mark, or a certified test report from a recognized independent testing laboratory acceptable to the Government indicating that the oil-burner, oil-pump, and electrical components offered have been tested and conform to the applicable requirements of UL Standard 296 and UL Standard 343, and applicable UL Standards respectively.

3.4 Materials (see 6.6) and components. Materials and components not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial ovens, provided the completed item complies with all provisions of this document.

3.4.1 Corrosion-resisting steel. Corrosion-resisting steel sheet shall conform to type 309 or 310 of ASTM A 167 or type 446 of ASTM A 176.

3.4.2 Carbon steel. Carbon steel sheets shall be hot rolled or cold-rolled. Sheet thickness shall be as shown in table I.

TABLE I. Carbon steel sheet thickness

Component	Minimum thickness
	Inch
Inner and outer shell	19 Gage (Nominal 0.0418)
Door, (including stiffeners)	17 Gage (Nominal 0.0538)
Compartment sides and top	19 Gage (Nominal 0.0418)
Compartment bottoms	10 Gage (Nominal 0.1345)

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3.4.3 Insulation. The heat insulating material for the outer shell (sides, back, top, and bottom) and doors, shall be mineral wool blanket or fiberglass. Insulation between plenum chamber, and lower compartment shall be mineral wool blanket. Loose fill insulation shall not be acceptable. The minimum thickness of the insulation shall be as specified in table II for the specific location. The complete lower front of the oven, both burner-side and fan-side, shall be insulated with 1/2-inch thick, minimum, solid asbestos insulation board. All spaces containing heat insulating material shall be sealed against entry of vapor, grease, and vermin.

TABLE II. Insulation thickness

Location	Minimum thickness
	Inches
Sides, back, top, and bottom	3
Doors	1-1/2
Between plenum chamber and lower compartment	1

3.5 Design and construction. The ovens shall be designed and constructed to minimize places where vermin might harbor or dirt collect, and to facilitate cleaning of surfaces where food particles or spillage might accumulate. Exposed bolts and nuts in such areas shall be kept to a minimum. The oven shall consist essentially of two compartments, each with a drop-down door. The compartments shall be indirectly heated by an oil-burner with heat exchanger and forced-draft air-recirculation fan. There shall be a separate, dampered hot-air supply to each compartment, with each damper controlled by a marked, adjustable quadrant, identified with the compartment it controls, on the face of the oven. The compartments shall reach 550°F within 1 hour after lighting the burner (see 4.5.3.1), shall maintain 400°F plus or minus 10°F for at least 1 hour (see 4.5.3.2) with a fuel consumption of not more than 25 cubic centimeters per minute (see 4.5.3.3). The oven shall satisfactorily bake a batch of bread within 1 hour (see 4.5.3.4) and shall recover to 450°F within 15 minutes after the baked bread is removed and the doors closed (see 4.5.3.4). Maximum overall dimensions and minimum compartment dimensions are shown on table III.

TABLE III. Oven dimensions

Dimensions	Overall (maximum)	Compartments (minimum)
	Inches	Inches
Height	76	14
Width	60	42
Depth	40	32

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* 3.5.1 Electrical characteristics. Unless otherwise specified (see 6.2), electrical characteristics of oil-burner, oil-pump, and air-circulation fan motors, and all safety controls, shall be 120-volt, 60 hertz, single phase. A transformer shall not be used to obtain the 120 volt characteristic of the safety control system. The electrical characteristics of the safety control system, as specified by UL Standard 296, are waived when a transformer would otherwise be required in order to obtain the 120 volt characteristic, except that all other requirements of UL Standard 296 for the safety control system shall be met.

3.5.2 Heating system. The heating system shall consist of an oil-burner, combustion chamber with domed top, heat exchanger, and flue; a plenum, air ducts, and recirculating fan. The fan-wheel shall be directly connected to the fan-motor shaft, and shall recirculate the heated air through the plenum chamber and oven compartments, with make-up air to compensate for minor leakage around compartment doors, flowing over the fan motor into the plenum chamber. The fan motor shall be equipped with oiling tubes with oil caps. The fan motor shall be completely outside the heat zone. The combustion chamber and plenum chamber shall be located below, and insulated from the oven compartments. The plenum chamber shall contain the combustion chamber and heat exchanger. The combustion chamber shall be lined with refractory brick. The design of the heating system shall permit replacement of brick without disturbing the combustion chamber, heat exchanger, or plenum chamber. The plenum chamber and heat exchanger shall be made of type 309 corrosion-resisting steel as specified in 3.4.1, at least 16 gage (nominal 0.0625) thick. The combustion chamber dome shall be made of type 309 corrosion-resisting steel at least 10 gage (nominal 0.1406) thick. Combustion chamber, ducts, and flue shall be made of any corrosion-resisting steel specified in 3.4.1. The flue material shall be at least 0.0565 thick. Products of combustion shall flow from the combustion chamber through the horizontal heat exchanger and vertical flue to a stove-pipe connection located on the top of the oven. The upper end of the vertical flue-duct shall be flush with the exterior top surface of the oven. An asbestos gasket shall be installed between the flue and the flue-opening in the oven top to provide a seal. The combustion chamber, heat exchanger, and flue-pipe shall not leak gases, oil fumes, or soot when tested as specified in 4.5.1. Secondary air shall be provided to the oil-burner in accordance with the manufacturer's recommendations. A 24-inch long external stove pipe extension shall be provided to fit the flue-opening.

3.5.3 Frame and inner and outer shell. The oven frame shall be fabricated from not less than 1-by 1-by 1/8-inch steel angles with corner and door posts formed from steel at least 0.1335 inch thick. The frame shall be welded. The inner and outer shells shall be riveted or welded. All sheet metal overlaps, including corners, shall be either welded or riveted. Four 5/16-inch diameter studs shall be located as shown on figure 1 on the front of the outer shell for the burner mounting plate.

3.5.3.1 Legs. Oven legs shall be adjustable and removable with a minimum of cross bracing and floor contact surface. All hollow sections shall be closed. Legs shall provide a minimum clearance of 6 inches between the floor and the lowest horizontal member. Legs shall be either corrosion-resisting steel or carbon steel.

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3.5.4 Compartments. The oven compartments shall be totally enclosed, with hot air inlet holes on the left side and return air holes on the right side. The interior surfaces of the oven compartments shall be readily accessible for cleaning and inspection. There shall be no exposed bolts, nuts, or threads within the oven compartments. All exposed, external corners and angles shall be closed and have a smooth surface.

3.5.4.1 Bottoms. The compartment bottoms shall be fabricated of carbon steel specified in 3.4.2, and all four edges bent 90 degrees to form a 1-inch flange. The bottom shall be supported by not less than 1-by 1-by 1/8-inch thick carbon steel angles, either welded to the frame, or bearing on members of equal strength which shall be welded to the frame.

3.5.4.2 Shelves. Each compartment shall be equipped with a removable horizontal shelf. Each shelf shall be capable of supporting a uniform load of 20 pounds per square foot of shelf surface without permanent distortion (see 4.5.3). Shelves shall be supported on steel channels welded to the compartment side-walls to divide the compartment into two equal sections.

3.5.4.3 Doors. The doors shall be welded assemblies of carbon steel (see 3.4.2), reinforced with stiffening members, or shall be welded steel frame and panel construction. Doors shall be flat on the inside surface and shall have rounded edges on the outer surface. Each door shall be hinged at the bottom so that, when open, it shall be level with, and form an extension of, the compartment bottom. The oven door hinges shall have means for lubrication. Doors shall fit snugly around the periphery of the compartment openings, and when in the open position shall support a load of 50 pounds per square foot of door surface without permanent deformation (see 4.5.4). The doors shall be counter-balanced or equipped with stops to stay open in the horizontal position. Each door shall have either a tube or bar-type handle.

3.5.4.4 Thermometer. Each oven shall be equipped with an angle-type thermometer for each compartment, with bulb filled with red fluid and mounted on the front of the oven. Each thermometer shall have a magnifying lens, protected by a metal cage, and black scale-markings on a white background graduated from 200° to 700°F. The spacing between 200° and 600°F shall be a minimum 5-1/2 inches.

3.5.5 Oil-burner and oil-pump.

* 3.5.5.1 The capacity of the oil-burner shall not exceed 1-1/2 gallons per hour of grade 2 fuel oil when tested as specified in 4.5.2. The burner shall be provided with a mounting plate having four 11/32-inch diameter holes spaced as shown on figure 1. The oil-burner shall be a horizontal, rotary cup type, with direct-connected motor and fan-rotated atomizing cup and nozzle. Fan drive by belt, chain, or gearing is prohibited, unless the acquisition document specifies other than 50 or 60 hertz current.

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The fuel shall be introduced through a tube within the hollow shaft of the motor or shall be introduced directly into the rear of the atomizing cup. Each burner assembly shall be provided with a strainer equipped with a removable screen or filter element. The burner shall be designed to burn a range of fuels meeting the following designations: grade DF-A (arctic grade), grade DF-1 (winter grade), and grade DF-2 (regular), diesel fuel oil conforming to VV-F-800; and grade 2 burner oil conforming to VV-F-815. Burners shall operate efficiently and without difficulty when tested as specified in 4.5.2. An observation port, designed to preclude warpage, shall be provided in the burner plate.

* 3.5.5.2 Oil pump. The oil pump shall be a separate unit for wall mounting and shall be of the rotary type, either gear, modified gear, or sliding vane type, equipped with a float for control of the motor. The float chamber shall be of such capacity that the pump shall be off at least one third of the time when delivering 3 gallons of fuel per hour (see 4.4.2.2). Means shall be provided to stop the pump motor automatically when the fuel supply fails. The float chamber shall be equipped with an overflow connection of sufficient size to allow the maximum fuel capacity of the pump, during continuous operation, to be returned by gravity to the fuel tank. The substitution of a second float switch, or other equipment, in lieu of a large overflow connection, shall not be allowed. The pump shall raise the fuel through 20 feet of vertical line and deliver not less than twice the maximum fuel rate of the burner when tested as specified in 4.4.2.2. The maximum temperature rise above room temperature of the pump motor winding coil insulation shall not exceed 180°F when operated continuously as specified in 4.4.1.2.

3.5.6 Valves and controls. The fuel control and combustion air control shall be interlocked to insure proper proportioning of air and fuel from full capacity to one-third this rate.

3.5.6.1 Solenoid oil valve. Each burner assembly shall have a manual resetting solenoid oil valve. The solenoid valve shall be installed on the burner unit between the burner hinge pin and the tubing. The solenoid valve shall automatically stop the flow of fuel to the burner upon failure of electric service. The flow of oil shall remain stopped until the valve is manually opened.

3.5.6.2 Draft regulator. Each oven shall have an automatic, adjustable draft regulator in the stove pipe (see 3.5.2).

3.5.6.3 Safety switch. The oil burner motor shall have an inclosed safety switch incorporating approved thermal overload protection. The switch shall open one side of two wire service to the motor and solenoid valve.

3.5.6.4 High-temperature limit switch. Each oven shall have a high temperature limit control switch which shall automatically shut off the oil burner when an air temperature of 700°F plus or minus 20°F is reached in the plenum chamber (see 4.5.6). The switch shall be installed on the lower right hand front of the oven and shall be inclosed in a box with a removable cover.

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3.5.6.5 Fan motor switch. Each oven shall have a control switch for the recirculating fan motor which shall be independent of the oil burner motor switch. The control switch shall be mounted next to the fan motor.

3.5.7 Bolted and riveted joints. Bolt and rivet holes shall be accurately punched or drilled, with burrs removed. Bolts shall have lock washers and nuts shall be drawn tight. Rivet heads shall be full, uniform in size, neatly made, concentric with holes, and in full contact with the surface of the members riveted. Speed nuts shall not be used anywhere on oven.

3.6 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. The combustion chamber, heat exchanger, and flue shall be fillet or butt welded, where required. Spot welding shall not be used for the combustion chamber, flue pipe, or heat exchanger.

3.7 Repair and maintenance. Provisions shall be made to allow for ready adjustment, service, and replacement of burner and pump without disturbing or disassembling of any component or equipment. In addition, there shall be free access to inspect or service and adjust any other component or equipment, as required, for proper operation of the complete ovens.

3.8 Marking for identification. Identification marking shall be on a photo-sensitive, anodized aluminum plate conforming to MIL-P-514, securely fastened to the exterior front of the oven. The plate markings shall be in accordance with MIL-STD-130.

3.9 Finish. The oven shall have the manufacturer's standard finish. The finish shall be smooth, uniform, and free of permanent soil or corrosion. Internal parts, including inside surfaces of inner and outer shells, shall be given a protective coating to resist corrosion.

3.10 Workmanship. The finished oven shall be free from marred surfaces, dents, punctures, fractures, rough edges, burrs, and slivers. All weld shall be smooth and uniform, completely fused, without slag, scale, or flux deposits, undercuts, cracks, burnholes, or fractures. All fasteners shall be sound and tight. No component shall be missing or malformed. All components shall be carefully assembled, properly adjusted and aligned.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the

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inspecions set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested as specified in 4.5.1 through 4.5.6. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Standards compliance. Proof of compliance with the UL requirements of 3.3 shall be submitted to the Government representative.

* 4.4.1.2 Pump operation test. The fuel pump shall be operated for a 1-hour period during which it shall continuously raise grade DF-2 diesel fuel oil through 20 feet of vertical pipe or tubing. The temperature of the oil passing through the fuel pump shall be maintained at 100°F ± 10°F during the test. The total flow in gallons during the 1-hour period shall be measured to within an accuracy of plus or minus 5 percent of the actual volumetric flow delivered through the vertical pipe or tubing. During the pump operation test, the maximum temperature rise above room temperature of the pump motor winding coil insulation shall be determined. Room temperature shall be 70°F ± 5°F, during this test. Temperature measurement shall conform to the requirements of UL 296, part 43. Any failure of the fuel pump to conform to the performance requirements of 3.5.5.2 during the test shall be considered a test failure. The lot shall consist of all fuel pumps produced essentially under the same conditions and offered for delivery at one time. The unit of sample shall be one complete fuel pump. The inspection level shall be level I and the acceptable quality level (AQL), expressed in terms of defects per hundred units shall be 4.0.

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* 4.4.2 In-process inspection. The contractor shall perform in-process inspections of all operations or assemblies which will be hidden from end item visual or dimensional examinations by further operations or assemblies during the manufacturing process. Objective evidence of the performance of in-process inspections and the results thereof shall be made available for review by the Government representative. The following operations and assemblies shall be included in this inspection:

a. Type of insulation and installation for conformance to 3.4.3.

b. Size of steel angles in oven frame (see 3.5.3).

c. Welding of frame compartment bottom support and door frames (see 3.5.3, 3.5.4.1, and 3.5.4.3).

d. Application of protective coating to inside surfaces of inner and outer shells (see 3.9).

* 4.4.3 End item visual examination. The end item shall be examined for the defects listed in table IV. The lot size shall be expressed in units of ovens. The sample unit shall be one completed oven. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE IV. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not smooth, uniform, or free of permanent soil and corrosion	X	
	Not manufacturer's standard finish		X
Design, construction, and workmanship (general)	Part missing	X	
	Part loose or not correctly adjusted and aligned		X
	Part fractured, split, punctured, torn, dented, creased, bowed, sprung, or malformed	X	

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Design, construction, and workmanship (general) (cont'd)	Part that will not function as intended	X	
	Adjustable part that cannot be adjusted	X	
	Rough edges, burrs, or slivers that could be injurious to personnel	X	
Dampers	Separate damper for each compartment not provided	X	
	Not controlled by marked quadrant, identified with the compartment, mounted on front of oven	X	
Fan	Not directly connected to motor	X	
	Motor not mounted outside heat zone	X	
	Fan and pump controlled by one switch	X	
	Not equipped with oiling tubes with flip-top oil caps		X
Combustion chamber	Not lined with refractory brick	X	
	Brick not replaceable without major disassembly	X	
Legs	Not 6-inch clearance under bottom of oven	X	
	Not adjustable or not removable		X
Compartments	Exposed bolts, nuts, or threads inside oven compartments	X	
Shelves	Not removable	X	
	Not in center of oven compartment		X
Doors	Not flat on inside or not rounded on outside		X
	Not hinged on bottom	X	
	When open, not level with oven bottom	X	
	Will not stay open	X	
	Handle not as specified		X
	No provisions for lubricating hinges	X	

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Oil-burner	Not rotary cup type	X	
	Observation port not as specified	X	
	Fuel control and combustion air control not interlocked	X	
Draft regulator	Does not function automatically	X	
	Not adjustable	X	
Controls	High temperature limit control not furnished	X	
	Not mounted in visible location		X
Insulation	Spaces not closed against dirt and vermin	X	
	Oven front not completely insulated	X	
Thermometer	Not installed in front of each compartment		X
	Bulb not filled with red fluid		X
	Not magnifying lens type		X
	Scale not white background with black markings		X
	Scale not graduated 200° to 700°F	X	
Fasteners	Not tight, broken, or not properly mounted	X	
Marking for identification	Missing, incomplete, not legible, or not specified type or size or material		X

* 4.4.4 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in this document. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of ovens. The sample unit shall be one oven. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing.

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* 4.4.5.1 Shelf load and door load testing. The end item shall be tested as specified in 4.5.4 and 4.5.5. The lot size shall be expressed in ovens. The sample unit shall be one oven. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.4.5.2 Operation and performance testing. The end item shall be tested as specified in 4.5.1, 4.5.2, 4.5.3, and 4.5.6. The lot size shall be expressed in ovens. The sample unit shall be one oven. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.6 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of MIL-R-11337.

4.5 Methods of inspection. All tests involving use of the burner shall be performed using diesel fuel conforming to grade DF-2 of VV-F-800.

* 4.5.1 Gas, fume, and soot leakage test. During the performance of each of the tests specified in 4.5.2, 4.5.3, and 4.5.6, the combustion chamber, heat exchanger, and flue shall be checked for leaking gases, oil fumes, and soot. Any evidence of nonconformance to the leakage requirement in 3.5.2 during performance of a test shall constitute failure of that test.

* 4.5.2 Burner operation test. With the burner in operating position, and ambient air temperature between 50°F and 100°F, the burner shall be operated and the fuel consumption rate shall be determined. Any failure to conform to the fuel consumption rate or burner operation requirements specified in 3.5.5.1 shall be considered a test failure.

* 4.5.3 Performance tests. The oven shall be subjected to the performance test specified in 4.5.3.1 through 4.5.3.4. In preparation for the performance tests, place thermocouples at the geometric center of each oven compartment and at the center of the back, top, bottom, and sides, 1-1/2 inches from the surfaces. A deflection type thermometer shall not be used. Any failure to conform to a performance requirement specified in 3.5 shall be considered a test failure. Temperature measurements shall conform to UL 296, part 43.

4.5.3.1 Heat-up test. With oven at room temperature (70°F ± 5°F, and the oven compartment doors closed, light the burner and record the time it takes the oven temperature to reach 550°F to determine conformance to the heat-up requirement to 3.5.

4.5.3.2 Heat maintenance test. With oven compartment temperature stabilized at 400°F for a minimum of 1-hour, check the temperature readings of the thermocouples for conformance to the heat maintenance requirement of 3.5.

4.5.3.3 Fuel consumption test. With the oven compartments temperature stabilized at 400°F and balanced by the adjustment of the damper control, check for conformance to the fuel consumption requirements of 3.5.

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4.5.3.4 Bake test. With the oven temperature at 450°F, load to capacity with 1-1/4 pound proofed dough pieces in pans. The pans shall be multiple-section type or single type, spaced to insure good circulation. Bake the dough at 360° to 370°F for about 1 hour. Remove the pans when the bread has reached a uniform brown color and hardness on all surfaces. Move the burner regulating valve lever to the high position and check the thermometers to determine compliance with the temperature recovery requirement of 3.5. Replace with a second full load of dough pieces and immediately close the door. Record the time required to bake the bread to determine compliance with the baking requirements of 3.5.

* 4.5.4 Shelf load test. Place the shelves in place in the oven compartments. Evenly distribute a load of 20 pounds per square foot over the shelves and allow to remain for 15 minutes. Remove load and check the shelves for distortion to determine conformance to 3.5.4.2. Any evidence of permanent distortion shall be considered a test failure.

4.5.5 Door load test. Open oven door fully, and evenly distribute a load of 50 pounds per square foot over the inner surface. After 2 minutes, remove the load and examine the door for permanent deformation to determine conformance to 3.5.4.3. Any evidence of permanent deformation shall be considered a test failure.

4.5.6 High temperature limit test. Place a thermocouple against the side of the combustion chamber dome. Open the oil regulating valve to its maximum position and allow the temperature to rise. Record the temperature at which the oil burner motor shuts down. Any nonconformance to the requirement in 3.5.6.4 shall be considered a test failure.

5. PACKAGING

5.1 Levels A, B, and C. Preservation, packaging, packing, and marking for each level of protection shall be in accordance with the applicable requirements of MIL-R-11337 (see 6.2).

6. NOTES

6.1 Intended use. The oven is intended for use in kitchens and bakeries for baking and roasting food stuffs.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.5).
c. Electrical characteristic, if other than specified (see 3.5.1).
d. Selection of the applicable levels of preservation, packaging, and packing (see 5.1).

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6.3 Definitions. The following definitions are applicable to this document:

Readily accessible - Readily accessible shall mean exposed without the use of tools for proper and thorough cleaning and visual inspection.

Closed - Spaces required to be closed shall have no openings large enough for the entrance of insects or rodents. An opening of 1/32 inch or less shall be considered closed.

6.4 Manuals. Any requirements for equipment manuals for the item covered by this document shall be included in DD Form 1423 Contract Data Requirements Lists, and cited in the contract.

6.5 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one complete oven. The contracting officer should include specific instructions in all acquisition documents, regarding arrangement for inspection and approval of the first article.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.4).

6.7 Marginal notations. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

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Navy - YD
Air Force - 84

Preparing activity:

Army - GL

Project No. 7310-0367

Review activity:

Army - MD

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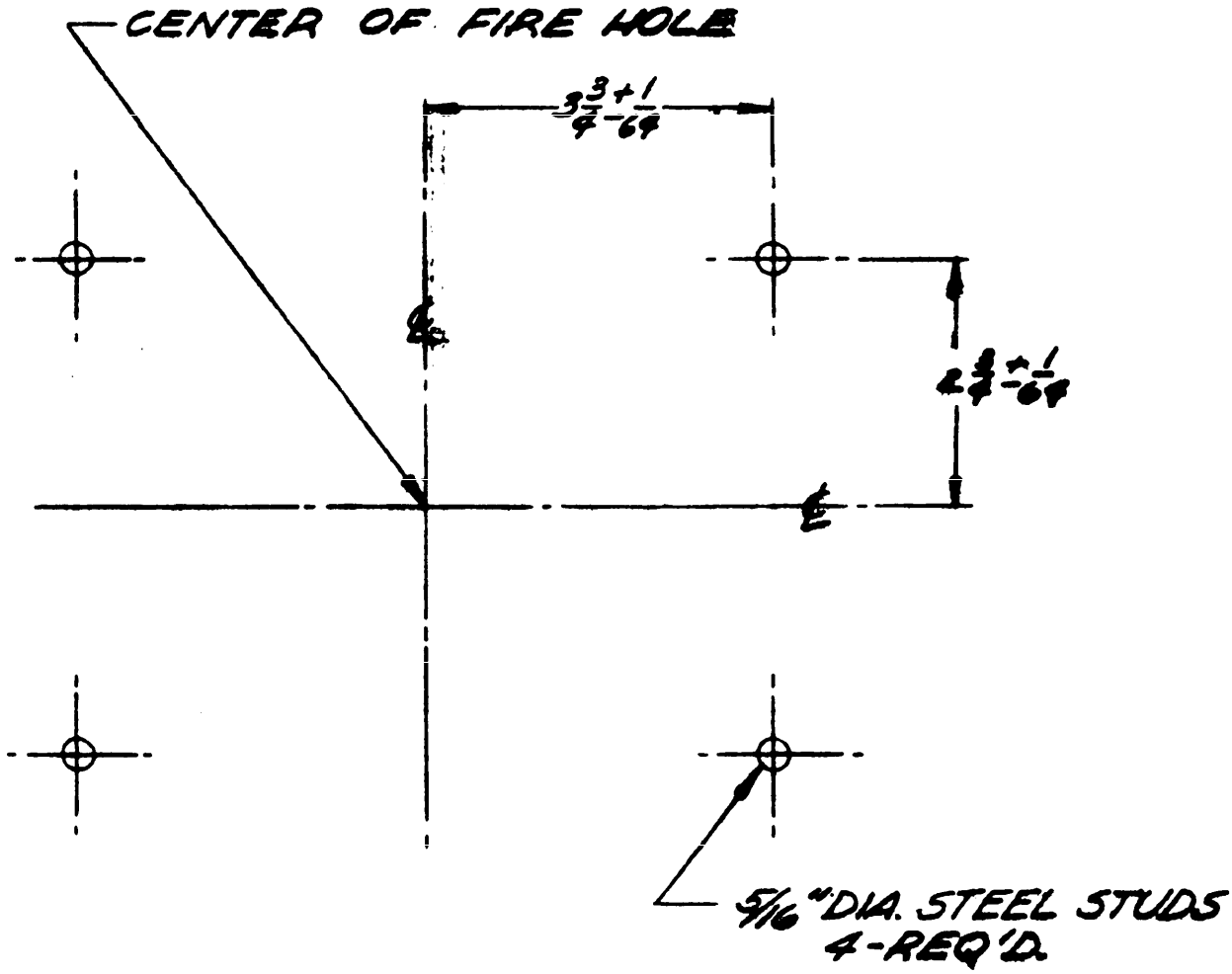


FIGURE 1. STUD LOCATION FOR MOUNTING BURNER PLATE.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-2478H	Oven, Baking And Roasting, Deck, Two-Compartment, Oil

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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